

The Corner Integrand for Rellich/Hadamard Boundary Integrals

A termwise Fourier–Bessel derivation, pure Dirichlet case

lappy development notes

Abstract

We derive, term by term, the integrand that a corner contributes to the Rellich normalization integral and to the Hadamard shape-derivative integral, for a pure Dirichlet eigenfunction. The derivation starts from the standard local-polar Fourier–Bessel expansion, expands the Bessel factor as its defining power series, and carries the expansion all the way to an explicit double series in powers of r — the singular corner exponents together with the smooth (“Bessel”) exponents that the existing derivation in `docs/corner_quadrature.tex` leaves in the abstract form $A_\mu(\lambda r^2)$. We then show how `lappy`’s node-placement and exactness substitutions act on this series, and why the Hadamard case needs a different substitution exponent than the Rellich case even though the two integrals share the same kernel $(\partial_n u)^2$.

Contents

1	The two identities, specialized to pure Dirichlet	1
2	The local Fourier–Bessel expansion	2
3	Expanding the Bessel factor termwise	3
3.1	The Bessel series	3
3.2	The normal derivative as an explicit double series	3
3.3	The squared normal derivative	4
3.4	The first few terms, worked out	4
4	The full boundary integrand, Rellich and Hadamard together	5
5	Rationalizing the integrand: the substitution	6
5.1	The naive substitution $t = r^\nu$	6
5.2	What <code>lappy</code> actually does: nodes from $t = r^\nu$, exactness from the weights	6
5.3	The general substituted Gauss–Jacobi rule	7
5.4	Why the Hadamard corner-moving weight needs $\text{sub} = 1/2$	7
6	Summary	8

1 The two identities, specialized to pure Dirichlet

For $-\Delta u = \lambda u$ in $\Omega \subset \mathbb{R}^2$ with a Zaremba boundary condition, the master Rellich–Pohozaev identity (proved by multiplying the PDE by $r \cdot \nabla u$ and integrating by parts; see `docs/boundary_quadrature_primer.te`

§2 for the full proof) is

$$\int_{\Omega} u v \, dA = \frac{1}{2\lambda} \oint_{\partial\Omega} (r \cdot N) [\partial_N u \partial_N v - \partial_T u \partial_T v + \lambda u v] \, ds + \frac{1}{2\lambda} \oint_{\partial\Omega} (r \cdot T) [\partial_T u \partial_N v + \partial_N u \partial_T v] \, ds,$$

with $r = x - x_0$ for an arbitrary reference point x_0 . On a *pure Dirichlet* domain, $u = v = 0$ on all of $\partial\Omega$, so every term carrying a bare u or v (rather than a normal derivative) vanishes, and so does every tangential derivative $\partial_T u, \partial_T v$ (the boundary values are identically zero, hence constant, along the boundary). What survives is a single term, and setting $v = u$,

$$\|u\|_{L^2(\Omega)}^2 = \frac{1}{2\lambda} \oint_{\partial\Omega} r_N \left(\frac{\partial u}{\partial n} \right)^2 \, ds, \quad r_N := (x - x_0) \cdot N. \quad (1)$$

This is the identity of `docs/corner_quadrature.tex` eq. (1) and `docs/rellich.md`'s I_1 term alone ($I_2 = I_3 = 0$ under pure Dirichlet, since both live on a Neumann arc that here does not exist). It holds for *every* $x_0 \in \mathbb{R}^2$; the choice of x_0 changes only how hard the boundary integral is to evaluate, not its value.

The companion identity is the Hadamard shape derivative of a simple Dirichlet eigenvalue under a boundary perturbation with normal velocity $V \cdot n$:

$$\frac{d\lambda}{d\varepsilon} = - \oint_{\partial\Omega} \left(\frac{\partial u}{\partial n} \right)^2 (V \cdot n) \, ds, \quad (2)$$

for u normalized in $L^2(\Omega)$ (`docs/orientation.tex` eq. 2, and the ‘‘Hadamard contract’’ promised to downstream code in `tests/test_shape_derivative.py`). Comparing (1) and (2): both are integrals of the *same* boundary quantity $(\partial_n u)^2$ against a weight that is linear in position relative to some reference. Taking $V = x - x_0$ (a uniform dilation of the boundary about x_0) turns (2) into $d\lambda/d\varepsilon = - \oint r_N (\partial_n u)^2 \, ds = -2\lambda \|u\|_{L^2(\Omega)}^2 = -2\lambda$, recovering (1) exactly (this is also the closed-form check $\lambda(\varepsilon) = \lambda/(1 + \varepsilon)^2$ for uniform scaling). So (1) is the special case of (2) with weight $V \cdot n = r_N$, and everything below that is derived for the kernel $(\partial_n u)^2$ applies to both; only the weight differs. This is why a single derivation of the corner integrand for $(\partial_n u)^2$ serves both formulas, and why §4 treats the weight as a free parameter.

2 The local Fourier–Bessel expansion

Place a corner of interior angle α at the origin, with its two edges along $\theta = 0$ and $\theta = \alpha$, and set

$$\nu := \frac{\pi}{\alpha}, \quad \nu_k := k\nu, \quad k = 1, 2, \dots$$

Every Dirichlet eigenfunction admits, in a neighborhood of the corner, the local expansion

$$u(r, \theta) = \sum_{k \geq 1} c_k J_{\nu_k}(\sqrt{\lambda} r) \sin(\nu_k \theta), \quad (3)$$

which vanishes identically on both edges by construction ($\sin(\nu_k \cdot 0) = 0$ and $\sin(\nu_k \alpha) = \sin(k\pi) = 0$). This is exactly `lappy`'s `FourierBesselBasis` convention in `lappy/bases.py`: a Dirichlet (`kind='sin'`) basis column evaluates as `jv(alphak_vec, sqrt(lam)*r) * sin(...)`, i.e. $J_{k\nu}(\sqrt{\lambda} r) \sin(k\nu \theta)$ termwise, with `alphak_vec` holding the sequence $k\nu$ for $k = 1, \dots$, order.

Remark 1 (A naming collision between this document and the code). The code’s local variable **alpha** is the sequence $\nu_k = k\nu$ used to call **ju**, not the interior angle. The interior angle is the code’s **phi**. In symbols: **phi** $\leftrightarrow \alpha$ (this document) and **alpha** $\leftrightarrow \nu_k$ (this document). We use the convention of `docs/corner_quadrature.tex` throughout: α is the interior angle, $\nu = \pi/\alpha$.

On the edge $\theta = 0$ the outward normal is $n = -e_\theta$, so, since $u \equiv 0$ along that edge, the only surviving derivative is the angular one:

$$\left. \frac{\partial u}{\partial n} \right|_{\theta=0} = -\frac{1}{r} \left. \frac{\partial u}{\partial \theta} \right|_{\theta=0} = -\sum_{k \geq 1} c_k \nu_k \frac{J_{\nu_k}(\sqrt{\lambda} r)}{r}. \quad (4)$$

The identical computation on $\theta = \alpha$ (where $n = +e_\theta$ / appropriately signed, since the corner sits at the wedge’s vertex and $\theta = \alpha$ is its other edge) gives the same series with $\cos(\nu_k \alpha) = (-1)^k$ in place of 1; we carry $\theta = 0$ throughout and note that the $\theta = \alpha$ edge differs only by that sign pattern, which does not affect any exponent or convergence statement below.

3 Expanding the Bessel factor termwise

This is the step that `docs/corner_quadrature.tex` leaves abstract (its eq. (4)–(5) invoke $J_\mu(z) = (z/2)^\mu A_\mu(z^2)$ with A_μ “entire,” without writing A_μ out). We now do so, to see explicitly which powers of r appear and with what coefficients.

3.1 The Bessel series

The Bessel function of the first kind has the convergent power series

$$J_\mu(z) = \sum_{m \geq 0} \frac{(-1)^m}{m! \Gamma(\mu + m + 1)} \left(\frac{z}{2}\right)^{\mu+2m} = \left(\frac{z}{2}\right)^\mu A_\mu(z^2), \quad A_\mu(w) := \sum_{m \geq 0} \frac{(-1)^m}{4^m m! \Gamma(\mu + m + 1)} w^m. \quad (5)$$

A_μ is an entire function of w (its radius of convergence is infinite, since the $m!$ in the denominator dominates any fixed growth in $\Gamma(\mu + m + 1)$), and $A_\mu(0) = 1/\Gamma(\mu + 1) \neq 0$ for $\mu > -1$, in particular for every $\mu = \nu_k > 0$ used here. Substituting $z = \sqrt{\lambda} r$ and $\mu = \nu_k$,

$$J_{\nu_k}(\sqrt{\lambda} r) = \left(\frac{\sqrt{\lambda} r}{2}\right)^{\nu_k} A_{\nu_k}(\lambda r^2) = \left(\frac{\sqrt{\lambda}}{2}\right)^{\nu_k} r^{\nu_k} \sum_{m \geq 0} a_{k,m} (\lambda r^2)^m, \quad a_{k,m} := \frac{(-1)^m}{4^m m! \Gamma(\nu_k + m + 1)}. \quad (6)$$

3.2 The normal derivative as an explicit double series

Insert (6) into (4). Each term contributes r^{ν_k-1} times a power series in r^2 , so

$$\left. \frac{\partial u}{\partial n} \right|_{\theta=0} = -\sum_{k \geq 1} \sum_{m \geq 0} c_k \nu_k \left(\frac{\sqrt{\lambda}}{2}\right)^{\nu_k} a_{k,m} \lambda^m r^{\nu_k-1+2m}. \quad (7)$$

Factor out the overall leading power $r^{\nu-1}$ (recall $\nu_k = k\nu$, so $\nu_k - 1 = (\nu - 1) + (k - 1)\nu$):

$$\left. \frac{\partial u}{\partial n} \right|_{\theta=0} = r^{\nu-1} F(r), \quad F(r) = \sum_{k \geq 1} \sum_{m \geq 0} b_{k,m} r^{(k-1)\nu+2m}, \quad b_{k,m} := -c_k \nu_k \left(\frac{\sqrt{\lambda}}{2}\right)^{\nu_k} a_{k,m}. \quad (8)$$

Two exponent families are visible directly in (8): the *corner* family $(k-1)\nu$, indexed by which radial mode k contributes, and the *smooth* (or “Bessel”) family $2m$, indexed by which term of the entire factor A_{ν_k} contributes. Every exponent in F is a nonnegative combination $j\nu + 2m$ with $j = k-1 \geq 0$ an integer and $m \geq 0$ an integer — F is *not* a power series in r alone unless ν is rational, but it is always a power series in the two formal variables r^ν and r^2 .

3.3 The squared normal derivative

Both (1) and (2) need $(\partial_n u)^2 = r^{2\nu-2} F(r)^2$. Write $G(r) := F(r)^2$; this is a Cauchy product of (8) with itself. Re-index the sum by the pair (p, q) where p collects all ways two corner-indices $j_1, j_2 \geq 0$ sum to p ($j_1 + j_2 = p$) and q collects all ways two smooth-indices $m_1, m_2 \geq 0$ sum to q ($m_1 + m_2 = q$):

$$\left(\frac{\partial u}{\partial n}\right)^2 \Big|_{\theta=0} = r^{2\nu-2} G(r), \quad G(r) = \sum_{p \geq 0} \sum_{q \geq 0} g_{pq} r^{p\nu+2q}, \quad g_{pq} = \sum_{\substack{j_1+j_2=p \\ m_1+m_2=q}} b_{j_1+1, m_1} b_{j_2+1, m_2}. \quad (9)$$

This is the explicit, finite-sum version of `docs/corner_quadrature.tex` eq. (6); each g_{pq} is a finite double sum (there are $p+1$ choices of (j_1, j_2) and $q+1$ choices of (m_1, m_2) , so $(p+1)(q+1)$ terms), each term an explicit product of two $b_{k,m}$ ’s, each of those an explicit product of a mode coefficient c_k , an elementary power of $\sqrt{\lambda}/2$, and a reciprocal-Gamma coefficient $a_{k,m}$ from (6). Nothing in g_{pq} is left abstract.

3.4 The first few terms, worked out

To make (9) concrete, write out the low-order terms of g_{pq} directly.

$(p, q) = (0, 0)$. Only $j_1 = j_2 = 0$ ($k = 1$ on both factors) and $m_1 = m_2 = 0$ contribute:

$$g_{00} = b_{1,0}^2 = c_1^2 \nu^2 \left(\frac{\sqrt{\lambda}}{2}\right)^{2\nu} \frac{1}{\Gamma(\nu+1)^2}.$$

This is the coefficient of the leading singular term $r^{2\nu-2}$ itself and depends only on the first mode’s amplitude c_1 .

$(p, q) = (0, 1)$. Still $j_1 = j_2 = 0$ (only $k = 1$), but now $(m_1, m_2) \in \{(0, 1), (1, 0)\}$:

$$g_{01} = 2 b_{1,0} b_{1,1} = -2 c_1^2 \nu^2 \lambda \left(\frac{\sqrt{\lambda}}{2}\right)^{2\nu} \frac{1}{4 \Gamma(\nu+1) \Gamma(\nu+2)}.$$

This is purely a smooth-family term ($p = 0$): it comes entirely from the next term of the entire factor $A_\nu(\lambda r^2)$ for the *same* mode $k = 1$, and contributes exponent $r^{2\nu}$ (i.e. $r^{2\nu-2} \cdot r^2$) to $(\partial_n u)^2$ — an even integer power of r times the overall singular prefactor.

$(p, q) = (1, 0)$. Now $(j_1, j_2) \in \{(0, 1), (1, 0)\}$, i.e. one factor from $k = 1$ and one from $k = 2$, both at $m = 0$:

$$g_{10} = 2 b_{1,0} b_{2,0} = -2 c_1 c_2 \nu \cdot 2\nu \left(\frac{\sqrt{\lambda}}{2}\right)^{\nu+2\nu} \frac{1}{\Gamma(\nu+1) \Gamma(2\nu+1)}.$$

This is purely a corner-family term ($q = 0$): it is the leading cross-term between modes $k = 1$ and $k = 2$, and it is the term that carries the *non-integer* exponent r^ν (on top of the $r^{2\nu-2}$ prefactor) whenever ν is irrational or has a large denominator — this is the term responsible for G failing to be analytic in r , and is exactly the obstruction that §5 exists to remove.

Remark 2 (Single-mode collapse). If only one mode is present ($c_k = 0$ for $k \geq 2$), then $F(r) = b_{1,-}$ -series only, i.e. $F(r) = (\text{const}) \cdot A_\nu(\lambda r^2)$ up to the constant absorbed in (8), and every corner-family index p in (9) beyond $p = 0$ vanishes: $G(r) = b_{1,-}$ -series squared = $\text{const} \cdot A_\nu(\lambda r^2)^2$, an analytic function of r^2 alone. This recovers exactly the special case noted in `docs/corner_quadrature.tex` §3.1: for a single mode, G is analytic and the corner integral is spectrally convergent under plain Gauss–Jacobi quadrature in r , with no substitution needed. The multi-mode cross-terms such as g_{10} above are precisely what breaks this for a general eigenfunction.

4 The full boundary integrand, Rellich and Hadamard together

On the straight edge $\theta = 0$, write the weight as a formal series in integer powers of r ,

$$W(r) = \sum_{m \geq 0} w_m r^m, \quad (10)$$

which covers both cases of interest:

- **Rellich.** $W(r) = r_N = (x - x_0) \cdot n$. On a straight edge this is an affine function of arclength; with x_0 placed anywhere, r_N restricted to $\theta = 0$ is the projection of $r e^{i0} - x_0$ onto $n = -e_\theta$, a degree-1 polynomial in r , so w_0, w_1 may be nonzero and $w_m = 0$ for $m \geq 2$. If x_0 is placed exactly at the corner, Proposition 1 below shows $W \equiv 0$ identically, collapsing the whole integrand to zero.
- **Hadamard.** $W(r) = V \cdot n$ for the perturbation’s normal velocity. A perturbation that leaves the corner fixed and only deforms the edge away from it typically vanishes to some even order at $r = 0$ ($w_0 = w_1 = 0$, generically $w_2 \neq 0$); a perturbation that *moves the corner itself* has $V \cdot n = O(r)$ with $w_1 \neq 0$, contributing an *odd* leading power.

Proposition 1 (Geometric escape at the corner). *If x_0 is placed at the corner (the origin), then $r_N \equiv 0$ on both straight edges emanating from it.*

Proof. On $\theta = 0$, $x - x_0 = r e^{i0}$ is parallel to the edge, while $n = -e_\theta$ is perpendicular to it, so $(x - x_0) \cdot n = 0$ pointwise; likewise on $\theta = \alpha$. $\square$

Combining (9) and (10), the full edge integrand is

$$W(r) \left(\frac{\partial u}{\partial n} \right)^2 \Big|_{\theta=0} = r^{2\nu-2} \underbrace{\left(\sum_{m \geq 0} w_m r^m \right) \left(\sum_{p,q \geq 0} g_{pq} r^{p\nu+2q} \right)}_{=: H(r)} = r^{2\nu-2} \sum_{p,q,m \geq 0} w_m g_{pq} r^{p\nu+2q+m}, \quad (11)$$

and the edge contribution to (1) or (2) is

$$I_{\text{edge}} = \int_0^R r^{2\nu-2} H(r) \, dr = \sum_{p,q,m} w_m g_{pq} \int_0^R r^{2\nu-2+p\nu+2q+m} \, dr, \quad (12)$$

convergent termwise for $\nu > 1/2$. Equation (11) makes explicit what `docs/orientation.tex` §3.2 states in table form: the integrand is always $r^{2\nu-2}$ times a series whose exponents are $p\nu + (2q + m)$ — the corner family $p\nu$ from the eigenfunction (unavoidable, fixed by the geometry) shifted by the *combined* smooth exponent $2q + m$, itself the sum of the eigenfunction’s own even smooth family $2q$ and the weight’s family m . For Rellich and most Hadamard weights $m \in \{0, 2, 4, \dots\}$ is even, so $2q + m$ stays even and the combined smooth family is unchanged from G alone; for a corner-moving Hadamard weight $m = 1, 3, \dots$ is odd, and $2q + m$ becomes *odd*, introducing an exponent family that G alone never has. This distinction is the entire content of §5.2.

5 Rationalizing the integrand: the substitution

Equation (12) is a sum of pure powers of r , but as a target for a fixed-order quadrature rule it is not directly tractable: the exponent set $\{p\nu + 2q + m\}$ is generically dense and irrational-stepped, so no polynomial rule is exact on it and Gauss–Legendre in r converges only algebraically. `lappy` evaluates I_{edge} (equivalently, the Cauchy data version of (9), formed numerically rather than from the coefficients above) by a change of variables that turns the corner exponents into integers.

5.1 The naive substitution $t = r^\nu$

Proposition 2 (Rationalizing substitution). *Under $t = r^\nu$ (equivalently $r = t^{1/\nu}$),*

$$\int_0^R r^\gamma r^{p\nu+2q} dr = \frac{1}{\nu} \int_0^{R^\nu} t^{(\gamma+1)/\nu-1} t^p t^{2q/\nu} dt, \quad \gamma := 2\nu - 2. \quad (13)$$

Proof. $dr = \frac{1}{\nu} t^{1/\nu-1} dt$ and $r^\gamma = t^{\gamma/\nu}$, so $r^\gamma r^{p\nu+2q} dr = \frac{1}{\nu} t^{\gamma/\nu+p+2q/\nu+1/\nu-1} dt$; collecting the exponent not multiplying p or $2q/\nu$ gives $\gamma/\nu + 1/\nu - 1 = (\gamma + 1)/\nu - 1$. $\square$

The corner-family exponent $p\nu$ becomes the *integer* p ; the outer singular prefactor r^γ becomes a Jacobi weight t^β with $\beta = (\gamma + 1)/\nu - 1 = 1 - 1/\nu$, admissible ($\beta > -1$) exactly for $\nu > 1/2$; but the smooth-family exponent $2q$ becomes $2q/\nu$, an integer only when $2/\nu \in \mathbb{Z}$. This is the caveat already present in `docs/corner_quadrature.tex` §3.1: among reentrant angles ($\nu < 1$), $2/\nu \in \mathbb{Z}$ happens only at $\alpha = 3\pi/2$ ($\nu = 2/3$, $2/\nu = 3$). For every other α , $t = r^\nu$ rationalizes the corner family completely but leaves the smooth family fractional — *high order* fractional (leading residual exponent $2/\nu > 2$), so Gauss–Jacobi quadrature in t converges at a high algebraic rate rather than spectrally, which is good enough in practice (machine precision by $n \approx 32$) but not exact at any finite order.

5.2 What `lappy` actually does: nodes from $t = r^\nu$, exactness from the weights

A further substitution $t = r^{1/q}$ (for $\nu = p/q$ in lowest terms) would rationalize the *full* exponent family $\{p\nu + m\}$ — including odd m , needed for a curved edge or a corner-moving Hadamard weight, see §4 — to integers $\{p \cdot p' + mq\}$ exactly, for any integer m . It is unusable as a node-placement substitution, however: writing $\tau = t^q$, the innermost quadrature node collapses as $\tau_{\min} \sim t_{\min}^q$, and for $q \geq 4$ this pushes nodes below `lappy`’s coordinate-collapse floor (`quad.KRESS_TAU_FLOOR` = 10^{-9}), where they round onto the corner itself once mapped through a segment’s parametrization — fatal, since the basis functions carry $1/r$ -type terms there.

`lappy`'s resolution (`lappy/quad.py`, `corner_rule_spec / cached_cornerjacgauss / cached_cornerinterpgauss`) separates the two jobs that a substitution was being asked to do:

- **Node placement** always uses the mild substitution $t = r^\nu$ (or more generally $t = \tau^{1/\text{sub}}$ for a chosen exponent `sub`, worked out below), whose crowding stays at $\tau_{\min} \sim 10^{-5}$ – 10^{-7} at the orders used in practice — far above the floor.
- **Exactness** is obtained instead from the *weights*. The exponent set $\{\gamma + j\nu + m\}$ appearing in (12) (relative to $R = 1$) is known in closed form, so every moment $\int_0^1 \tau^e d\tau = (e + 1)^{-1}$ is known exactly; a quadrature rule with `sub` = ν -placed nodes τ_i and free weights w_i is made exact on that exponent set by solving the linear system

$$\sum_i w_i \tau_i^e = \frac{1}{e + 1}, \quad e \in \{\gamma + j\nu + m : 0 \leq j \leq j_{\max}, 0 \leq m \leq m_{\max}\}, \quad (14)$$

using *fewer* exponents than nodes (a minimum-norm least-squares solve), which keeps $\sum_i |w_i| \approx 1$ and avoids the ill-conditioning ($\text{cond} \sim 10^{13}$ – 10^{19}) of the square system. This is `cached_cornerinterpgauss`, and it is exact on the full family $\{\gamma + j\nu + m\}$ including odd m , without requiring ν to be rational at all.

5.3 The general substituted Gauss–Jacobi rule

The rule underlying both `cornerjac` and the node placement of `cornerinterp` is a direct generalization of Proposition 2 to an arbitrary substitution exponent `sub`, matching `lappy.quad.cached_cornerjacgauss`. For an integrand $\tau^\gamma \times (\text{series in } \tau^{\text{sub}})$ on $[0, 1]$, substitute $\tau = t^{1/\text{sub}}$:

$$d\tau = \frac{1}{\text{sub}} t^{1/\text{sub}-1} dt, \quad \tau^\gamma = t^{\gamma/\text{sub}},$$

so

$$\int_0^1 \tau^\gamma f(\tau) d\tau = \int_0^1 t^\beta \tilde{f}(t) dt, \quad \beta = \frac{\gamma + 1}{\text{sub}} - 1, \quad (15)$$

where $\tilde{f}(t) := \frac{1}{\text{sub}} t^{1/\text{sub}-1} f(t^{1/\text{sub}})$ absorbs the Jacobian; a Gauss–Jacobi rule (t_i, W_i) for weight t^β on $[0, 1]$ then gives nodes and weights

$$\tau_i = t_i^{1/\text{sub}}, \quad w_i = \frac{W_i}{\text{sub}} t_i^{1/\text{sub}-1}, \quad (16)$$

exact whenever f is a polynomial in τ^{sub} of degree $\leq 2n - 1$ (in t). Setting `sub` = ν and $\gamma = 2\nu - 2$ recovers exactly (13)–Proposition 2 above ($\beta = 1 - 1/\nu$); `cached_cornerjacgauss` implements (16) for general `sub`, and this is the same formula used for the alternate substitution below.

5.4 Why the Hadamard corner-moving weight needs `sub` = 1/2

Take `sub` = ν (the default, exact for the eigenfunction's own corner family) and consider a Hadamard weight term $w_m r^m$ with m a fixed positive integer (§4: $m = 1$ for a corner-moving perturbation). By (15), $r^m \mapsto t^{m/\nu}$: since `sub` = ν is chosen to rationalize the eigenfunction's exponents $p\nu \mapsto p$, it does the *opposite* to an externally-imposed integer power m — it sends an already-integer exponent to a fractional one, m/ν , and worse, a *small* one (since $\nu \in (1/2, \infty)$) generically makes m/ν close to m but not equal to it). The residual fractional part is what the

rule cannot integrate exactly, and Gauss–Jacobi’s convergence rate on a term with residual order δ near the endpoint is only algebraic, $n^{-(2\delta+2)}$ — so a small residual is the *worst* case, not a negligible one. Measured on a 1.5π sector against 40-digit truth (`docs/orientation.tex` §3.2), the weight r^1 term alone costs four digits at order 32 relative to r^0 or r^2 :

	$p = 0$	$p = 1$	$p = 2$	$p = 3$
sub = ν , order 32	2.9×10^{-14}	4.6×10^{-7}	8.5×10^{-14}	4.0×10^{-14}

Switching to sub = 1/2 reverses which exponents land on integers: by (15), $r^m \mapsto t^{2m}$ (exact, an integer for *every* integer m , independent of ν), while the eigenfunction’s own corner family $r^{p\nu} \mapsto t^{2p\nu}$ becomes fractional in general — but its exponents now grow by 2ν per term, spaced widely enough (unlike the sub = ν residual above, which is a small offset from an integer) that Gauss resolves the resulting smooth tail at once. The same table under sub = 1/2 at order 16:

	$p = 0$	$p = 1$	$p = 2$	$p = 3$
sub = 1/2, order 16	4.7×10^{-15}	1.2×10^{-14}	1.1×10^{-14}	1.0×10^{-14}

— uniformly at machine precision, at half the order, and with *no* assumption that ν is rational (verified in `lappy` at $\nu = 1/1.37$ and $\nu = 1/\varphi$ to 10^{-15}). This is `lappy`’s `boundary_quadrature(..., weight_family='integer')` (implemented in `lappy/eigfun_integrals.py`, forcing `kind='cornerjac'`, sub = 1/2 at every singular corner): it exists specifically because the weight in (2) is, for a corner-moving shape perturbation, of a different exponent family than the weight r_N in (1) ever is on a straight edge, even though both integrals share the identical kernel $r^{2\nu-2}G(r)$ derived in §3. The default remains sub = ν (`weight_family='even'`) because it is equally accurate and cheaper for every weight `lappy` evaluates internally, all of which are even in m .

6 Summary

weight $W(r)$	exponent family m	combined family	sub	rule
Rellich, straight edge, x_0 at corner	—	$W \equiv 0$, no integral	—	(none needed)
Rellich, straight edge, general x_0	$\{0, 1\}$	$p\nu + 2q + \{0, 1\}$	ν	<code>cornerjac/cornerinterp</code>
Hadamard, corner-fixed velocity	even	$p\nu + 2q + 2q'$	ν	<code>cornerjac/cornerinterp</code>
Hadamard, corner-moving velocity	odd (from r^1)	$p\nu + 2q + \{1, 3, \dots\}$	1/2	<code>cornerjac, weight_family='inte</code>

In every row the singular kernel is the same object, $(\partial_n u)^2 = r^{2\nu-2}G(r)$ derived termwise in (9); only the weight’s own exponent parity determines which substitution exponent sub rationalizes the product, and hence which `lappy` rule is selected.